

1 • Chemistry Foundations

UNIT 1 STUDY LIST

1.0 Describing Matter

I can:

- ☐ Describe matter as collections of discrete particles.
- ☐ Explain a change using the law of conservation of mass.

Reading Guide

Read section 1.2-1.3

1.1 Measurements

I can:

- ☐ Appropriately read and record measurements with a digit of uncertainty.
- ☐ Identify the significant digits in a measurement.
- ☐ Write a number in both standard decimal notation and scientific notation.
- ☐ Carry out arithmetic operations, expressing the result with the correct number of significant figures.
- ☐ Describe the accuracy (using percent error) or precision (using the average deviation) of a set of measurements.

Reading Guide

Read sections 1.4-1.6

1. What are the SI units for mass? Length? Volume? Time? Amount of substance?
2. How do you enter " 1.23×10^{-4} " into your scientific calculator? Try it.

3. What is the conversion factor between kg and g? (Write " $\square \text{ kg} = \square \text{ g}$ ".) Between kg and mg? Between nm and m? Between L and cm^3 ? Between mL and cm^3 ?
4. Define *precision* and *accuracy*.
5. Does the *average deviation* describe a set of measurement's precision or accuracy? Why? Does the *percent error* describe a value's precision or accuracy? Why?
6. Write the formula for percent error.
7. When a measurement contains a zero, when is the zero a significant figure?
8. What is the rule for reporting the sum or difference of numbers?
9. A battery plate is 94.0% lead by mass. Express this percent as a fraction (include appropriate units).
10. What must you do first when you are adding or subtracting two numbers in scientific notation?

1.2 Dimensional Analysis

I can:

- ☐ Identify appropriate units for a measurement.
- ☐ Relate the volume of a liquid (in mL) to the volume of the container (in cm^3).
- ☐ Explain the meaning of metric prefixes. ("King Henry Died ... drinking chocolate milk")
- ☐ Use dimensional analysis to convert like units. (e.g. length $\rightarrow$ length)
- ☐ Use dimensional analysis to convert units for area and volume (e.g. $\text{cm}^2 \rightarrow \text{cm}^2$).
- ☐ Use dimensional analysis with unlike relationships (ratios and rates) to convert unlike units.

1.3 Density

I can:

- ☐ Describe and compare objects' mass, volume, and density quantitatively, graphically, with particulate diagrams, and from macroscopic descriptions (such as whether objects will sink or float).
- ☐ Use particulate descriptions to explain the differences of the density in solids, liquids, and gases

Reading Guide

Read section 1.4

1. What is the mathematical formula for *density*?
What are the units for density?
2. What is the density of pure water at 25°C? Why is it important to state the temperature?
3. Suppose you have a sample of ice and liquid water at 0°C. Why does the ice float? Use numerical values to justify your explanation.

4 • Atoms and Elements

STUDY GUIDE

4.1 Atoms, Ions, Isotopes

Lesson Objectives

I can:

- ☐ describe the models of the atom proposed by Dalton, Thomson, and Rutherford, and explain the limitations and/or inaccuracies of each.
- ☐ describe the composition of an atom and the features (relative mass, size, charge, location) of subatomic particles in an atom written in isotopic notation.
- ☐ explain the relationship between two atoms with different protons, neutrons, or electrons.
- ☐ provide evidence for the existence of isotopes using mass spectrometry data.
- ☐ write the average atomic mass of an element and explain that it represents the average mass of naturally occurring isotopes.
- ☐ calculate the average atomic mass of an element using abundance data of isotopes.

Reading Guide

Read sections 2.1-2.3

1. In one sentence, describe the main contribution by each of the following scientists in understanding the atom:
(a) Democritus, (b) John Dalton, (c) Sir Joseph John Thomson, (d) Ernest Rutherford.
2. State postulates of John Dalton's atomic theory, and the laws of conservation of matter, constant composition, and multiple proportions.
3. For the proton, electron, and neutron, describe their (1) location in the atom, (2) mass, (3) charge.
4. Benjamin Franklin discovered what facts about electric charges?
5. List and describe the three types of radiation (e.g. their charge, deflection)
6. Define the following: atomic number, atomic mass unit (amu), mass number, isotope, atomic mass.
7. For the atom, $^{23}_{11}\text{Na}$, state the number of protons and neutrons.

4.2 The Periodic Table

Lesson Objectives

I can:

- ☐ describe the organization of the periodic table in its order and structure (incr protons, families/periods), its categories (metals/nonmetals; alkali/ alkaline earth/ transition/ inner transition/ halogens/ noble gases).
- ☐ describe typical properties of metals and non-metals.
- ☐ identify the common elements by name, symbol and position on the periodic table.
- ☐ describe properties of elements in standard state, including the diatomic elements and which elements are gases (noble gases + N_2 , O_2 , F_2 , Cl_2), liquids (Br_2 and Hg), and solids (the rest).

Reading Guide

Read sections 2.5

1. In the periodic table, vertical columns are called _____ and horizontal rows are called _____.
2. Describe five macroscopic properties of metals. Compare descriptions of metals and non-metals.
3. What are two "missing" elements predicted by Dmitri Mendeleev?
4. State the "law of chemical periodicity."
5. State three facts about elements that belong to: (1) Alkali Metals: Group 1A, (2) Alkaline Earth Metals: Group 2A, (3) Transition Metals, (4) Group 3A, (5) Group 4A, (6) Group 5A, (7) Group 6A, (8) Halogens, (9) Noble Gases.

